Homework 1 – Direct kinematics

Provided Material

1. “SimScene” folder: containing the robot model in the simulator
2. “SimLibraryFunctions” folder: containing some functions to interact with the simulator.
3. File MAIN_homework1.m: script where the simulated robot is positioned according to a given joint configuration.

Setup

The setup, shown in Figure 1, is composed of a Reachy 2 robot by Pollen Robotics (https://docs.pollen-robotics.com/hardware-guide/specifications/general/). We will focus on the left arm only which is composed of 7 revolut joints.

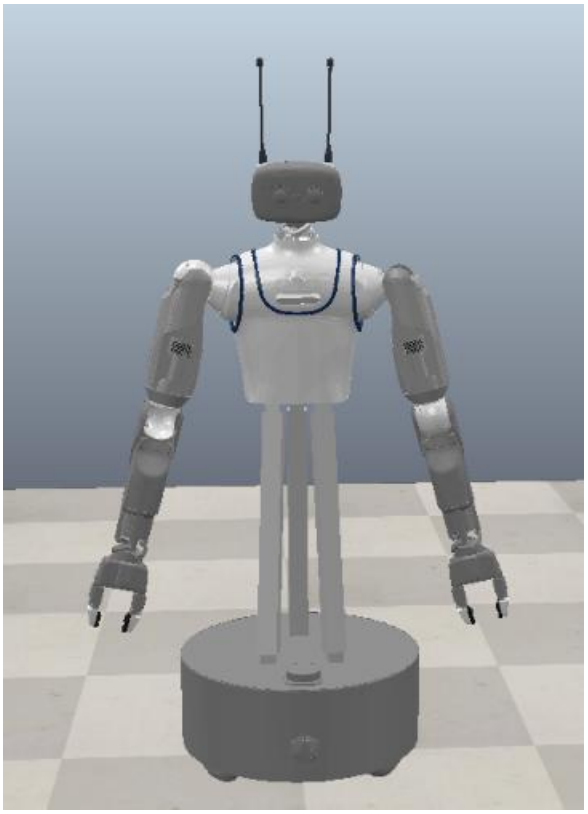


Figura 1 – Robot in CoppeliaSim

The following joint limits exist for the robot left arm:

1	$-\pi$	π	[rad]
2	$-\pi/2$	$\pi/2$	[rad]
3	-4.7124	$\pi/2$	[rad]
4	-3.8208	-1.4708	[rad]
5	-0.35	0.35	[rad]
6	1.2208	1.9208	[rad]
7	$-\pi/2$	$\pi/2$	[rad]

Figure 2 shows relevant lengths of the robot (in mm).

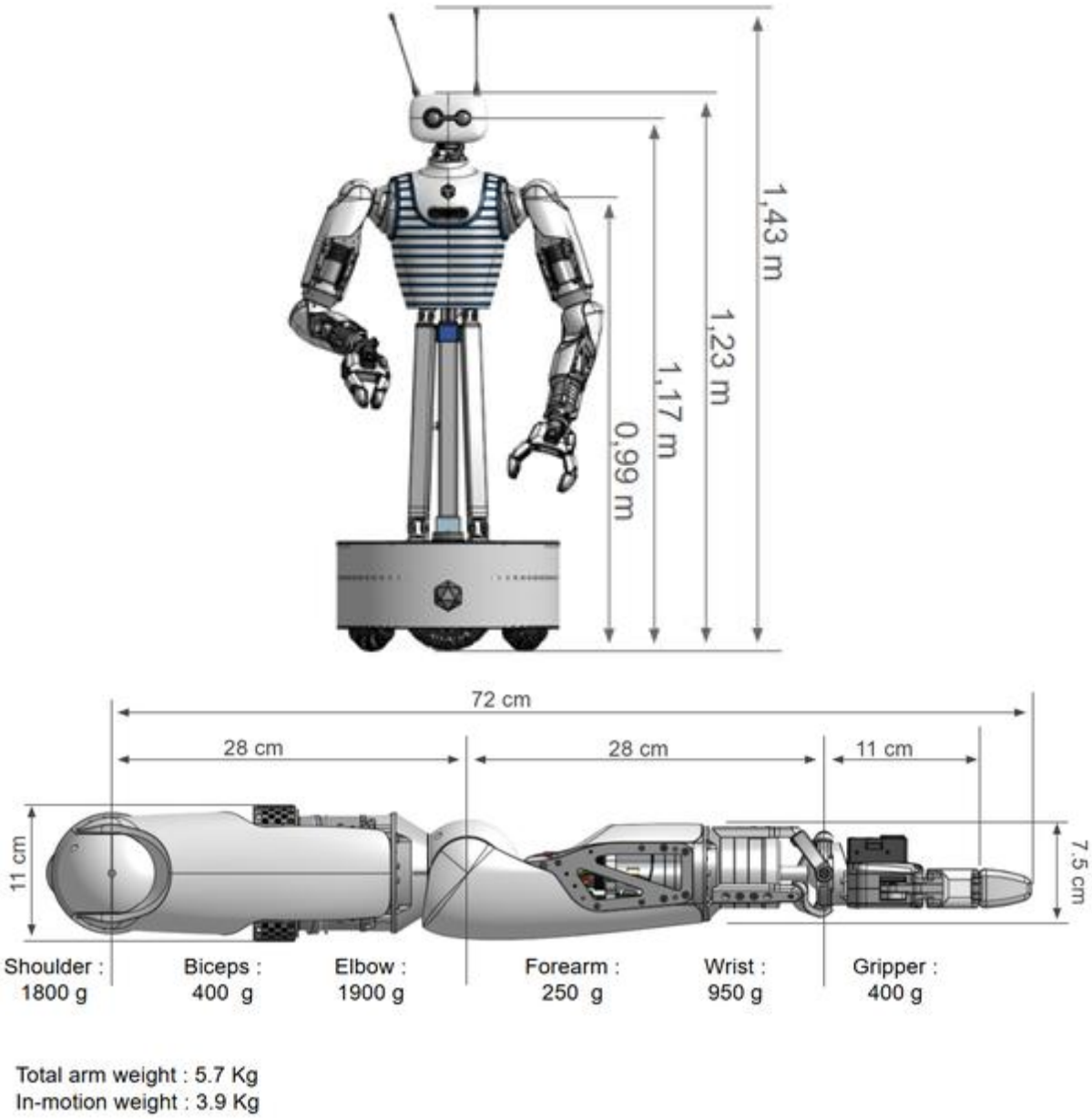


Figure 2 – Dimensions of the robot

Information relative to the kinematics

Base frame

The base frame coincides with the simulator one and is shown in Figure 3.

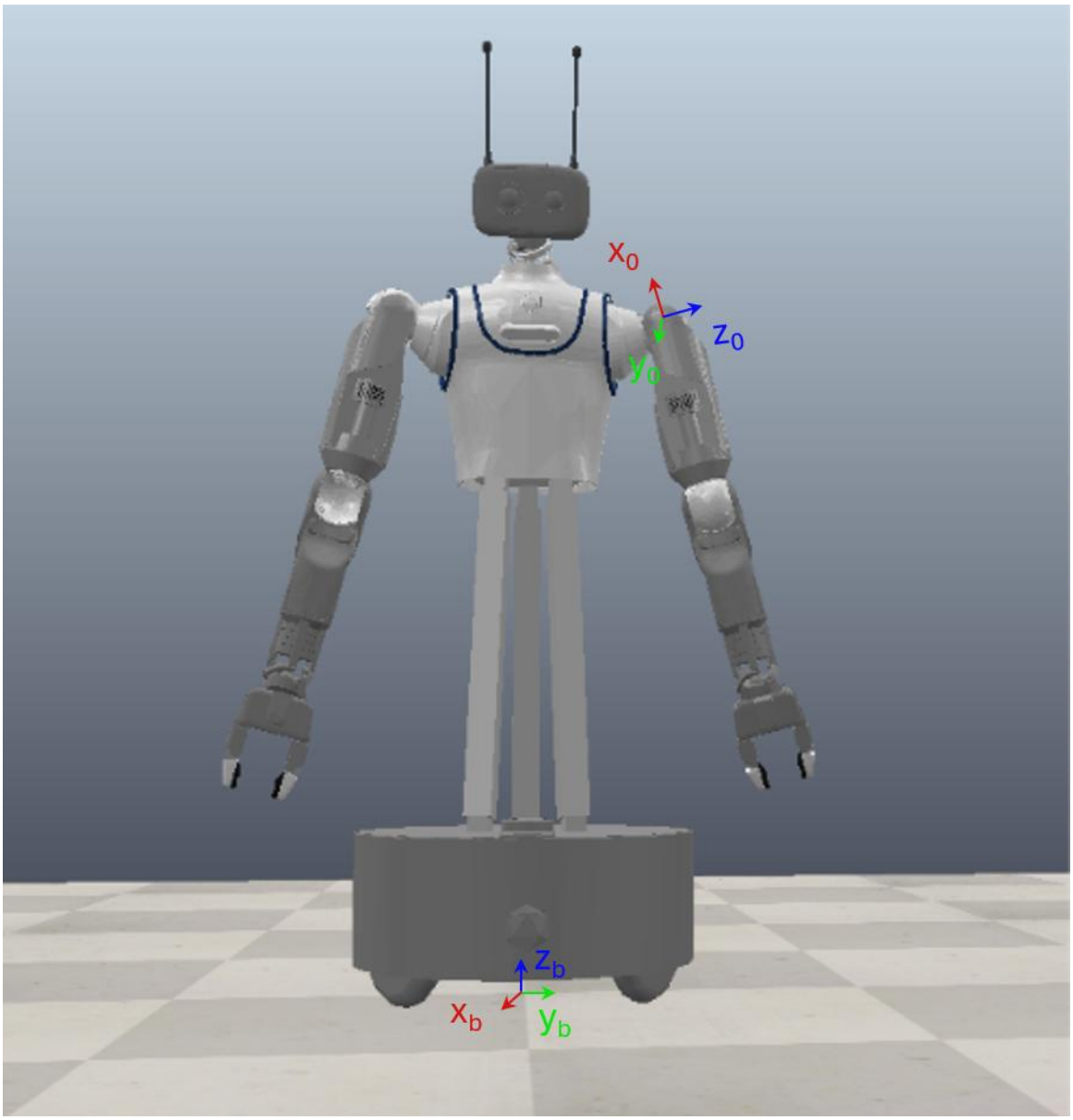


Figure 3 – Representation of the base frame and the 0 frame

The 0 frame of the robot is set as shown in Figure 3. Specifically, considering the rotations and translations depicted in Figures 2 and 4, the transformation matrix A_0^b is defined as follows

$$A_0^b = \begin{bmatrix} -0.0449 & 0.9848 & 0.1677 & 0.0155 \\ -0.2549 & -0.1736 & 0.9513 & 0.2000 \\ 0.9659 & 0 & 0.2588 & 0.9960 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

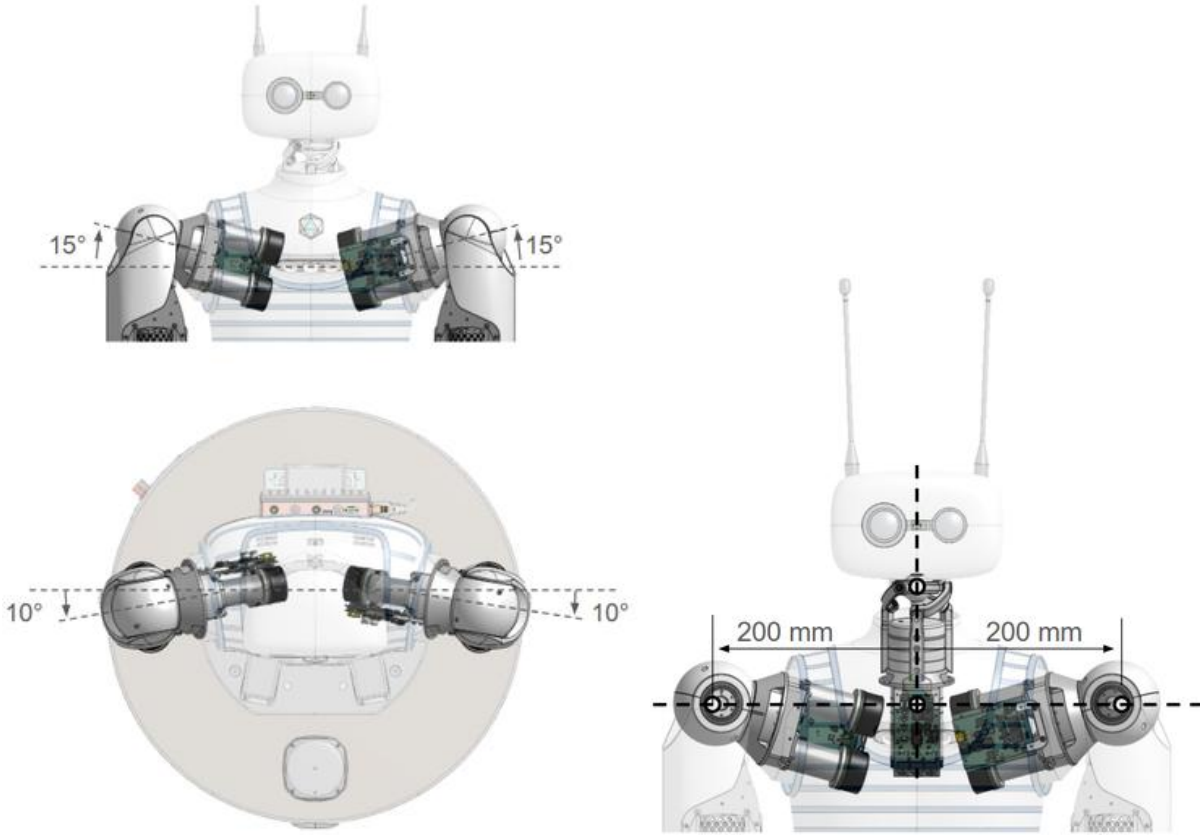


Figure 4 - Rotations and translations for 0 frame

DH frames

The DH frames should be defined in such a way that, with the following joint vector

$$q = \left[0, -\frac{\pi}{2}, -\frac{\pi}{2}, -\frac{\pi}{2}, 0, \frac{\pi}{2}, 0\right]^T \quad (rad\ rad\ rad\ rad\ rad\ rad\ rad),$$

the manipulator is configured as in Figure 3 (or equivalently Figure 1).

To define the z axis of each frame, it is recommended to consider the rotation direction of each joint (highlighted in the next figure).

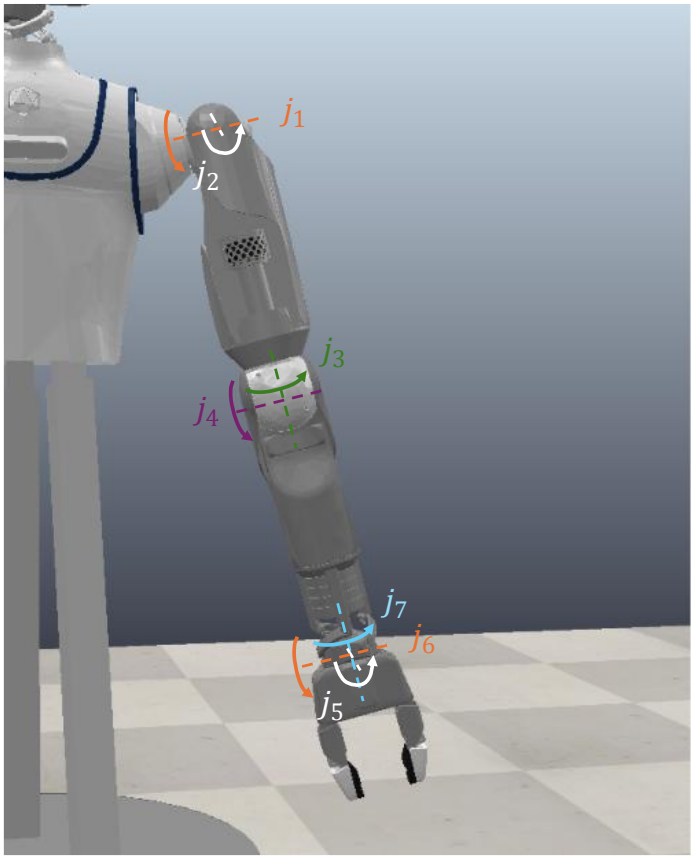


Figure 5 – Rotations of the joints

End-effector frame

The robot is equipped with a parallel gripper as depicted in Figure 6. Place the end-effector frame as depicted in the picture (this is frame "l_EE_FRAME" in CoppeliaSim).

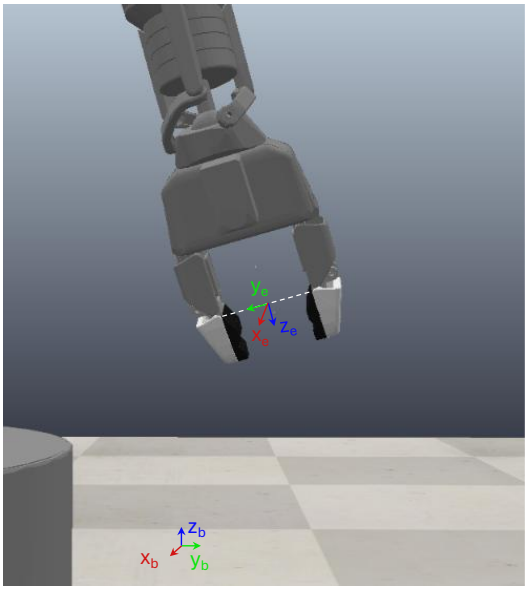


Figure 6 - End Effector frame

Notes for connecting via Matlab to the simulator

The file GUIDE_SIMULATOR.txt contains notes for using the simulator with MATLAB.

In particular, the following files need to be visible in the MATLAB path:
`remoteApiProto.m`
`remApi.m`
`"remoteApi.dll" (Windows), "remoteApi.dylib" (Mac) or "remoteApi.so" (Linux)`
These files are in the simulator installation folder.

In addition, the file remoteApiConnection.txt (in the simulator installation folder) should be modified as follows:

portIndex1_port = 19998

portIndex1_debug = true

portIndex1_syncSimTrigger = true

Note: there is an offset between the joint variables expressed according to the DH convention and the joint variables in CoppeliaSim. In particular, it holds

$$q_{coppelia} = q_{DH} + \left[0, \frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{2}, 0, -\frac{\pi}{2}, 0\right]$$

This is already considered in the function "moveLeftArm" in SimProjectUtils

What to deliver

a) All the code produced, ready to be executed, appropriately commented and with corresponding instructions for use. In particular, the code should contain a function that, given the joint variables, returns the position and orientation of the end effector. The function should allow specifying as input which Euler angles to use for orientation (ZYZ or RPY) and should return:
   a. The position of the end-effector frame (in meters)
   b. The orientation of the end-effector frame according to the Euler angles specified in the input (in radians)
   c. The rotation matrix relative to the end-effector frame
   d. The quaternion vector associated with the rotation matrix.

   e.g. [p, phi, R, quat] = direct_kin(q, 'ZYZ');

b) A report written independently by each student. The report should detail the direct kinematics of the robot showing the chosen frames (and giving reasons for the choices) in accordance with the DH convention and the corresponding table. The report should not include theoretical aspects of the book.